DEVARSHI AGGARWAL

Robotics & Physical AI Engineer | Robot Learning • Isaac Sim / Isaac Lab • RL / IL • Computer Vision

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SUMMARY

Robotics / Physical AI Engineer with 4+ years of experience in deep learning and computer vision, currently developing simulation-first robotic manipulation systems using NVIDIA Isaac Sim and Isaac Lab. Hands-on experience in reinforcement learning, imitation learning, teleoperation-based data collection, robotic task/environment development, and vision-based policy training. Built and evaluated learning-based control pipelines for dual-arm manipulation and developed VR teleoperation systems for collecting demonstrations for IL/VLA training. Strong background in PyTorch, computer vision, scalable AI deployment, and experimental evaluation. Currently pursuing an M.Tech in Artificial Intelligence & Machine Learning from BITS Pilani.

EXPERIENCE

Samsung Electro Mechanics, India
Robotics Engineer | Deep Learning Engineer *December 2024 - Present*

- Develop simulation-based **Physical AI and robotic manipulation pipelines** for dual-arm robotic systems using **NVIDIA Isaac Sim and Isaac Lab**, including environment construction, task definition, observations, actions, rewards/terminations, and policy evaluation.
- Built and evaluated **reinforcement learning and imitation learning** policies for robotic manipulation tasks; improved task success rate from **0% to 99%** through systematic experimentation across 100+ training runs.
- Developed a **Meta Quest 2 VR teleoperation pipeline** for collecting robotic demonstrations in simulation for imitation-learning and Vision-Language-Action training workflows.
- Built end-to-end robotics data pipelines covering **teleoperation, demonstration collection, dataset preparation, policy training, rollout evaluation, and failure analysis**.
- Worked on **vision-based and multimodal robotic learning**, integrating RGB observations and robot proprioception for learning-based control.
- Previously developed perception systems for driver monitoring including gaze estimation, drowsiness detection, face recognition, and facial landmark detection with >90% accuracy.

Jio, India
Deep Learning Engineer *April 2022 - December 2024*

- Improved Jio ANPR detection accuracy from **90%** → **95%** and recognition accuracy from **83%** → **94%**, achieving ~99% end-to-end accuracy after tracking.
- Designed scalable real-time computer-vision pipelines using **NVIDIA DeepStream**, increasing deployment density from **8 to 108 camera streams per GPU**.
- Developed production AI systems for person analytics, face recognition and vector-search-based access control using PyTorch/OpenCV/Milvus.

Agro Informatics Lab, IIT Bombay, India
Computer Vision Intern *Aug 2021 - March 2022*

- Extracted differentiating features from images of various genotypes of rice.
- Used machine learning techniques to identify the best species for inter-species breeding.
- Made tools and scripts for data labelling and data augmentations which greatly improved manual labelling time by 75%.

Freelancer
Freelancer *July 2020-July 2021*

- Worked for various clients on small and medium-sized projects using python and deep learning skills.
- Developed a deep reinforcement learning model for trading on the metatrader tool, for an independent client.

Department of Computer Science, IIT Bombay
Machine Learning Intern *July 2019- May 2020*

- Developed a computer-vision/OCR pipeline for structured information extraction from handwritten and printed forms.
- Built annotation tooling that improved dataset preparation workflows.

Publications

Leaf Count Aided Novel Framework for Rice (Oryza sativa L.) Genotypes Discrimination in Phenomics: Leveraging Computer Vision and Deep Learning Applications

On 3-Degree 4-Chordal Graphs IEEE CICT 2020

EDUCATION

BITS – Pilani, India *July 2025 – Present*
Masters of Engineering, Artificial Intelligence & Machine Learning;
- Relevant coursework: Deep Reinforcement Learning, Deep Neural Networks, Machine Learning, AI & ML for Robotics, 3D Computer Vision / relevant current electives.

IIITDM-Kancheepuram, India *July 2015 – June 2019*
Bachelor of Engineering, Computer Science; Cumulative GPA: 8.18/10.0
- Relevant coursework: Artificial Intelligence Search Methods for Problem Solving, Pattern Recognition, Digital Image Processing, Joy of Computing using Python, Operating Systems, Probability Theory, Linear Algebra, Differential Equations, Calculus, Design and Analysis of Algorithms, Database Systems

Sunbeam School Lahartara – Varanas**i,** India *July 2012-June 2014*
Secondary School; Cumulative GPA(CBSE): 93.0%/100.0

SKILLS

Robotics & Physical AI: NVIDIA Isaac Sim, Isaac Lab, PhysX, Robotic Manipulation, Robot Learning, Reinforcement Learning, Imitation Learning, ACT-style Visuomotor Policies, Residual Policy Learning, Teleoperation, Simulation Evaluation, USD, URDF
Deep Learning & Vision: PyTorch, OpenCV, Ultralytics, DeepStream, Object Detection, Tracking, Representation Learning
Programming: Python, SQL, Shell, C, Rust (Beginner)
ML & Infrastructure: Docker, NumPy, Pandas, scikit-learn, Milvus
Engineering: CAD Design, 3D Printing

EXTERNAL LINKS

- GitHub - https://github.com/devarshi16
- Portfolio Website - https://devarshi16.github.io/

OTHER ACHIEVEMENTS

- Secured **AIR-5320 (GEN) in IIT JEE-Advanced 2015**